

InterTrack: A Behavior World Model for Robust Humanoid Whole-Body Tracking with Environment Interaction

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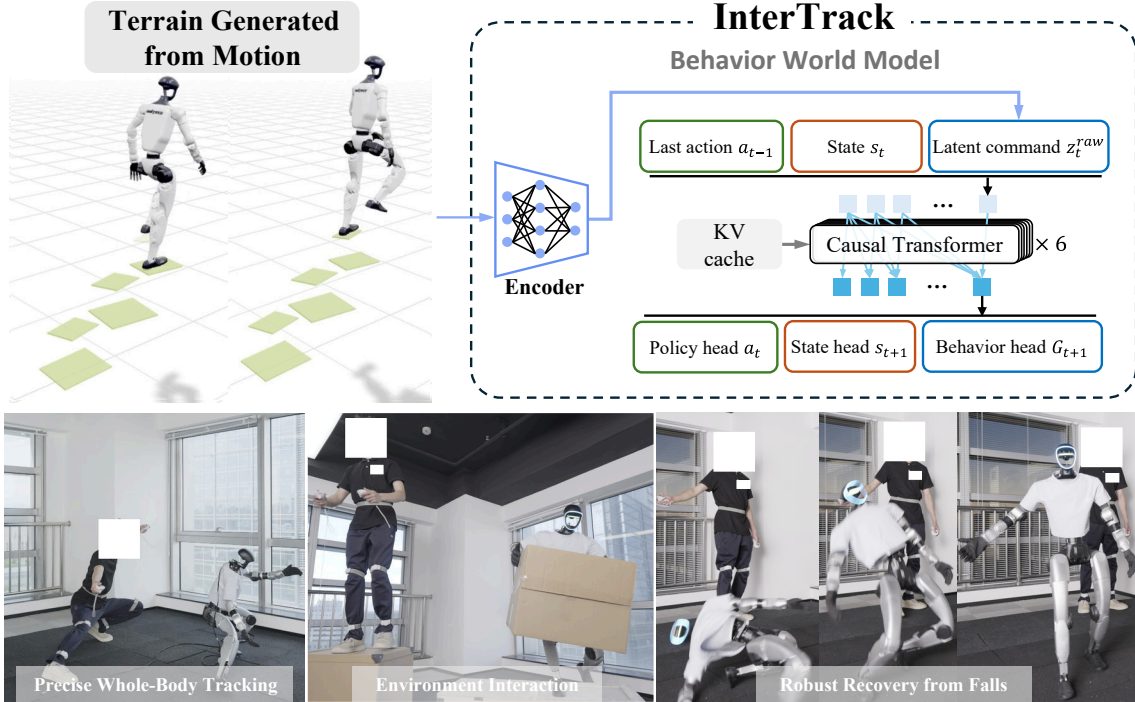


Fig. 1. **InterTrack learns environment-conditioned whole-body tracking.** A behavior world model predicts future states and behaviors, thereby capturing environment-conditioned dynamics; a unified policy supports precise tracking, environment interaction, and fall recovery under live whole-body commands.

Abstract—General-purpose motion trackers enable humanoid robots to follow diverse whole-body motions while maintaining balance, but are trained only on flat ground, failing to exploit bipedal mobility over complex terrain. Cross-terrain controllers, meanwhile, are task-specific or accept only low-dimensional locomotion commands. We introduce InterTrack, the first behavior world model (BWM) for robust whole-body tracking with environment interaction. Its Transformer jointly predicts the next action, state, and behavior distribution, learning environment-conditioned dynamics. To scale interaction training data, an automatic annotation pipeline reconstructs 3D support geometry from retargeted motions. At deployment, the policy handles commands implausible in the current environment in a “best-effort” manner. Quantitatively, InterTrack achieves an 81.3% success rate on terrain interaction (4.3× the best evaluated baseline) and a 99.3% fall-recovery rate, while also improving free-space tracking and outperforming three leading tracking baselines across all of these regimes. To our knowledge, we provide the first demonstration of real-time cross-terrain whole-body teleoperation on a humanoid robot, alongside object interaction, stable responses to missing supports, and robust recovery from falls. Details and videos: <https://intertrack2026.github.io>.

I. INTRODUCTION

General-purpose motion tracking turns a humanoid into a general control interface. A single policy can imitate

diverse human motions while handling low-level balance and physical feasibility, allowing a human operator [1], planner, or vision-language-action model [2] to provide coarse motion intent instead of precise torque-level commands. Recent trackers scale this recipe to hundreds or thousands of hours of retargeted motion [3]–[8], enabling agile teleoperation and data collection [1].

The advantage of humanoid legs lies in traversing and interacting with complex environments, yet existing trackers are typically trained only on flat ground in empty scenes. A general-purpose tracker trained only in this setting therefore never learns that a contact surface can change the robot’s dynamics or enable or preclude a commanded behavior: a chair makes it possible to sit down but prevents stepping backward, while a payload shifts the center of mass and requires the policy to adapt in order to maintain balance. Policies that do support rich interaction or locomotion across varied terrain are either specialized for particular skills [9], [10] or limited to directional velocity commands for locomotion, precluding general manipulation tasks [11]. Training interaction at motion-corpus scale requires pairing each motion with consistent scene geometry, which ordinary motion-capture datasets cannot provide. SceneBot [12] reconstructs terrain

post hoc but uses 2.5D elevation maps and per-link contact prompts; height fields cannot represent overhangs or vertical supports, and the required contact labels are unavailable from a live operator.

Environment interaction also compounds the robustness problem. On flat ground, prior work simply broadens the reference corpus until many commands become learned and executable. With varied geometry, feasibility is conditional: sitting down is meaningful on a chair but not in free space, and a level-ground gait may be viable on the floor but not on stairs. Moreover, simple fallbacks become inadequate: emergency stops or standing-pose recovery can not only interrupt the task but also create safety risks when the robot is performing a dynamic task on complex terrain. A useful controller should therefore preserve as much of the command as its current situation permits. It should also remain controllable after a disturbance or fall, rather than interrupting the task to invoke a separate recovery policy.

We introduce InterTrack, a *behavior world model* for robust humanoid whole-body tracking with environment interaction. Its causal Transformer emits actions and predicts the next proprioceptive state and the conditional distribution over the next latent behavior command, learning how the current environment affects robot dynamics and which behaviors it permits. At deployment, the filter uses this distribution to project implausible commands toward compatible behaviors, producing a stable “best-effort” response that retains intent. Our automatic pipeline reconstructs 3D support geometry from contact evidence in retargeted trajectories, providing paired motion–geometry data for interaction training at motion-corpus scale. In simulation, InterTrack reaches 81.3% terrain-interaction success ($4.3\times$ the best baseline) and 99.3% fall recovery, with the best overall performance on standard free-space tracking. Hardware trials show live whole-body teleoperation across terrain, object interaction, missing-support robustness, and disturbance recovery.

Our contributions are:

- a causal behavior world model that jointly predicts actions, future states, and future behavior distributions for environment-conditioned whole-body tracking;
- an automatic annotation pipeline that reconstructs 3D support geometry from retargeted motions, scaling interaction training beyond scene-captured datasets;
- stable “best-effort” behavior under implausible commands and recovery from falls within a unified policy.

II. RELATED WORK

a) Whole-body tracking and behavior foundation models: Physics-based imitation [18] made motion capture a scalable source of dense control supervision. Modern systems expand motion diversity with quantized latent commands [3], sparse mixture-of-experts Transformers [4], expert distillation [5], diffusion or masked completion [8], [19], and heterogeneous command interfaces [20]. Other work emphasizes agile skills [21], perturbation robustness [7], or the fidelity–diversity trade-off [22], [23]. Teleoperation systems [1] expose these policies as real-time interfaces. Behavior

TABLE I
CAPABILITY COMPARISON, REPRESENTATIVE METHODS.

	Diverse whole-body tracking	Low-latency teleop.	Terrain inter-action	Object inter-action	Robust to implausible commands	Robust to falls
GMT [6]	✓	✗	✗	✗	✗	✗
SONIC [3]	✓	✓	✗	✗	✗	✗
Humanoid-GPT [5]	✓	✓	✗	✓	✗	✗
PHP [13]	✗	✗	✓	✗	✗	✗
OmniRetarget [9]	✗	✗	✓	✓	✗	✗
CMP [14]	✓	✓	✗	✓	✓	✗
BFM-Zero [15]	✓	✓	✗	✗	✗	✓
SceneBot [12]	✓	✗	✓	✓	✗	✗
Perceptive BFM [16]	✓	✗ [‡]	✓	✗	✗	✓
PGMT [17]	✓	✗ [†]	✓	✗	✗	✓
InterTrack (ours)	✓	✓	✓	✓	✓	✓

[†]PGMT postdates InterTrack’s public release; its demo shows over 1 s of teleoperation delay, exceeding our low-latency criterion. [‡]Perceptive BFM’s 21-frame future window implies a look-ahead delay of at least 0.4 s at 50 Hz.

foundation models extend shared pretraining across control modes [15]. Most whole-body trackers are restricted to flat ground. The few exceptions—Perceptive BFM [16], PGMT [17], and SceneBot [12]—respectively require at least 0.4 s of future-reference buffering, exhibit over 1 s of teleoperation delay in demonstrations, and rely on per-link contact prompts unavailable from live motion capture. InterTrack scales up interaction data approximately tenfold while supporting real-time cross-terrain whole-body teleoperation.

b) Interaction with terrain and objects: Classical contact planning and model-predictive control achieve structured locomotion [24]; learned perceptive policies extend this to stairs, slopes, constrained terrain, and parkour [11], [13]. Scene-aware retargeting and data generation support large-object or whole-body interaction [9], [10], while video reconstruction and terrain-aware generation couple motions to geometry [25]. These systems are skill-specific or accept only directional velocity commands, providing joystick-level locomotion insufficient for complex manipulation. InterTrack instead controls terrain and object interaction from whole-body motion commands.

c) Robustness to commands and falls: Domain randomization and other perturbations widen a tracker’s training distribution [7], [21], while learned priors constrain behavior to a motion manifold [23]. In legged manipulation, CMP [14] shows that long-horizon command safety reduces to a per-step test and that the distributions of training and feasible commands closely match. We extend both insights to humanoid whole-body tracking, where command feasibility also depends on terrain and contacts. Fall recovery is typically delegated to specialized standing or protective policies [26]; BFM-Zero [15] shows that it can be integrated into a general policy. InterTrack further unifies fall recovery, environment interaction, and agile teleoperation in one tracker.

III. METHOD

A. Problem Setting

We control a Unitree G1 through joint-level PD targets. At step t , the policy maps proprioception s_t , the previous target

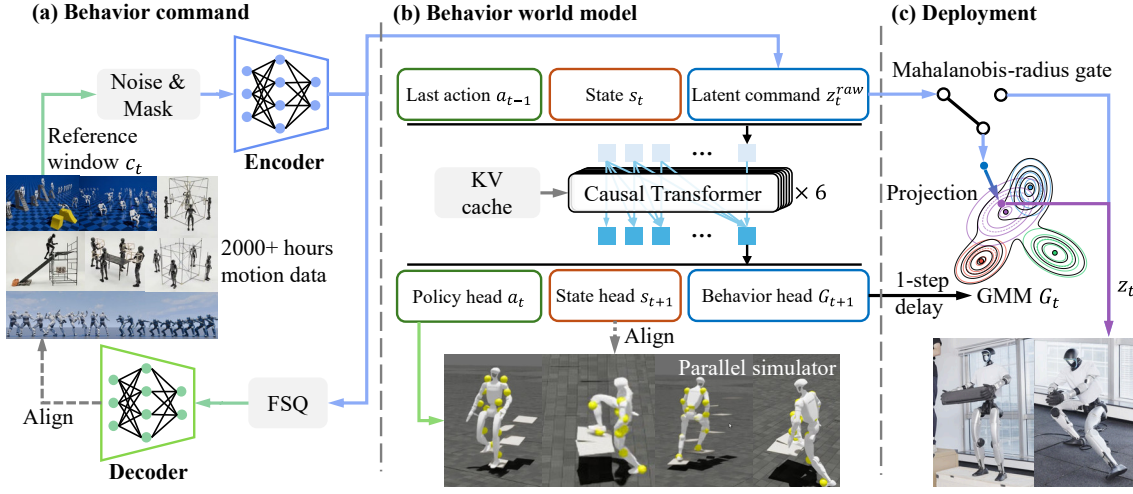


Fig. 2. **InterTrack overview.** A partially masked reference window is encoded into a latent behavior command; a finite scalar quantizer (FSQ) regularizes this latent through a training-only reconstruction branch. The causal Transformer consumes the command, proprioception, and previous action, then jointly predicts the action, next state, and command distribution. At deployment, the preceding distribution identifies and corrects implausible commands.

a_{t-1} , and a short reference window c_t to the next target a_t ; Appendix A specifies the exact interface. The reference comes from teleoperation or a high-level policy.

Like other general-purpose trackers, ours deliberately omits vision, following a brain–cerebellum division of labor: an upstream system perceives the scene and selects the reference, while the compact, high-rate controller focuses on dynamics. Uncontacted geometry cannot affect the current dynamics; once contact occurs, its effect is exposed more directly through proprioception than vision. Thus, when terrain and objects make execution depend on supports, contacts, and loads, the controller infers these conditional dynamics from proprioceptive history without scene observations or privileged contact labels.

B. Behavior World Model

To make the controller understand and adapt to environment-conditioned dynamics, we train it as a behavior world model rather than an action predictor alone (Figure 2). Alongside the action, it predicts the robot’s next proprioceptive state and the conditional distribution of the next behavior command. Predicting the next state teaches the controller how current contacts change the robot’s dynamics, while predicting the next command teaches it which behaviors remain compatible with the current physical situation. Together, these objectives let the controller account for interaction dynamics and behavior feasibility when choosing an action.

A reference encoder first maps the reference window c_t to a continuous raw behavior command $z_t^{\text{raw}} \in \mathbb{R}^{d_z}$, where $d_z = 64$; its unchanged or corrected version z_t conditions the controller. The core input $e_t = [s_t, a_{t-1}, z_t]$ is then processed by a six-layer causal Transformer [27] with rotary embeddings [28]. Each layer attends over 32 frames, yielding a 187-frame effective receptive field that captures long-horizon interaction history. A per-layer KV cache accelerates autoregressive inference. From the Transformer output at step t , three heads produce

$$a_t \in \mathbb{R}^{29}, \hat{s}_{t+1} \in \mathbb{R}^{67}, \mathcal{G}_{t+1} = \{\pi_k, \mu_k, \log \sigma_k\}_{k=1}^4, \quad (1)$$

where π_k are the predicted mixture weights and $\mathcal{G}_{t+1}(z) = \sum_{k=1}^4 \pi_k \mathcal{N}(z; \mu_k, \text{diag}(\sigma_k^2))$. This standard diagonal mixture allows distinct behavior modes to be represented without conflating them in one broad Gaussian.

Because both the command predictor and deployment filter operate in the latent space, its geometry directly determines whether the mixture can represent and delimit distinct behavior modes. We therefore attach a training-only FSQ [29] branch to the reference encoder. As in SONIC [3], this auxiliary branch decodes the quantized command back to the clean reference and re-encodes the reconstruction for cycle-consistency supervision. Without quantization, minimizing the cycle-consistency MSE can favor a shrinking latent scale, pulling encoded behaviors progressively closer together. FSQ prevents this shortcut: if the scale collapses, nearby behaviors fall into the same quantization bins and can no longer retain enough information for reconstruction, while the finite number of levels also limits how much information can be folded into any one latent dimension. Unlike SONIC, however, our main Transformer branch bypasses FSQ and always consumes the continuous command, preserving fine motion variations at deployment instead of exposing control to discrete token transitions.

The reference encoder must also reconcile long-horizon terrain alignment with hardware observability. Our 187-frame receptive field calls for long episodes: frequently resetting them to bound global drift would truncate the history the model is designed to exploit, whereas leaving drift uncorrected eventually misaligns the reference with its paired terrain. The robot-relative root-translation input provides this correction but requires a global root position unavailable on hardware. We therefore expose it intermittently during training, teaching the policy to re-anchor when it is present and rely on interaction history when it is absent; it remains masked throughout deployment.

The PPO [30] objective is

$$\mathcal{L} = \mathcal{L}_{\text{PPO}} + \lambda_{\text{recon}} \mathcal{L}_{\text{recon}} + \lambda_{\text{cycle}} \mathcal{L}_{\text{cycle}} + \lambda_{\text{state}} \|\hat{s}_{t+1} - s_{t+1}\|^2 - \lambda_{\text{beh}} d_z^{-1} \log \mathcal{G}_{t+1}(z_{t+1}), \quad (2)$$

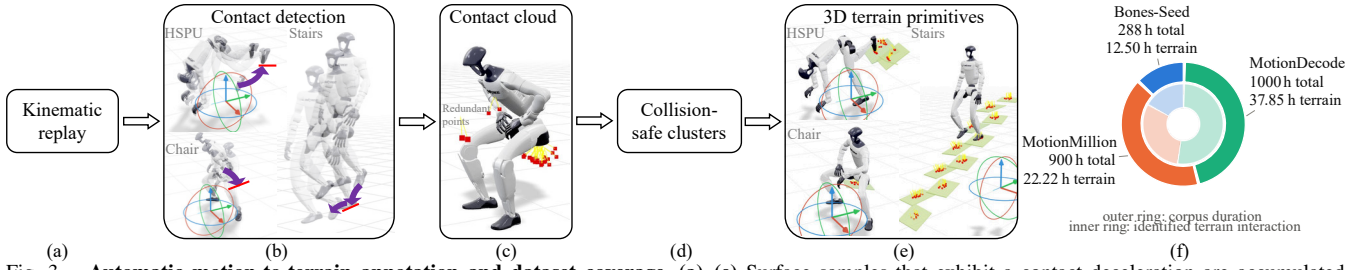


Fig. 3. **Automatic motion-to-terrain annotation and dataset coverage.** (a)–(e) Surface samples that exhibit a contact deceleration are accumulated, filtered against whole-body penetration, clustered, and fitted with spatial collision primitives. (f) Outer and inner rings show the total duration of each source corpus and its identified terrain-interaction duration, respectively.

Loss weights, optimization, reward, curriculum, and randomization details are in Appendix B.

C. Automatic Spatial Terrain Annotation

Contact-conditioned training requires motion–geometry pairs, but existing routes scale poorly. MeshMimic [25] reconstructs motions and scenes from video, inheriting video-quality limits; OmniRetarget [9] relies on prebuilt scenes, leaving diversity asset-bound. We instead infer support geometry from retargeted trajectories (Figure 3(a)–(e)), utilizing the large scale of existing motion-capture corpora. SceneBot [12] takes a similar approach but reconstructs 2.5D height fields, whereas our full 3D geometry can capture free space beneath structures such as tables.

For each retargeted trajectory, we uniformly sample the collision surfaces of contact-relevant links and replay the motion kinematically in MuJoCo [31], recording world-space positions and outward normals. A Savitzky–Golay filter [32] suppresses numerical jitter before a per-sample state machine detects contact: low normal and tangential speeds must coincide with recent sharp normal deceleration, distinguishing support from momentarily slow motion. Contact ends upon excessive displacement or sliding. Detected contacts are accumulated into a point cloud.

We discard candidates intersecting the robot at any replay frame, preventing unwanted collisions. An affinity graph links survivors with aligned normals and small normal-direction offsets. Density-based spatial clustering of applications with noise (DBSCAN) [33] separates distinct supports, including adjacent surfaces at different heights. Tangent-plane principal component analysis (PCA) estimates each cluster’s principal directions and uses geometric primitives to approximate the point cloud. Clusters whose bounding boxes contain large empty regions are recursively split along the principal axis to fit non-convex or annular structures. Finally, a penetration-safe binary search expands each primitive laterally to tolerate tracking errors.

We apply the pipeline to three retargeted corpora (Figure 3(f)): Bones-Seed [34] (288 h), MotionMillion [35] (900 h), and MotionDecode [36] (1000 h), identifying 12.50, 22.22, and 37.85 hours of terrain interaction. The resulting terrain primitives are instantiated as static bodies in Isaac Lab [37] and paired with their corresponding reference motions during training. We evaluate the accuracy of the resulting annotations in Appendix D.

D. Robustness to Implausible Commands and Falls

Command feasibility depends on the environment: a command that is valid on certain terrain may be physically implausible on flat ground, making blind execution unsafe. Simple solutions, such as an emergency stop or recovery to a standing pose, interrupt the task and may themselves be unsafe while the robot is traversing complex terrain. Following CMP [14], we use a per-step command-distribution test as a practical proxy for command feasibility. With paired terrain–motion training data, we flag commands that are out of distribution (OOD) under the behavior distribution conditioned on the current environment, even if the same motion appears elsewhere in the dataset. The preceding prediction $\mathcal{G}_t$, formed before observing c_t , defines this test from the current physical history (Figure 2): $c_t \rightarrow E \rightarrow z_t^{\text{raw}} \xrightarrow{\mathcal{G}_t} z_t^* \rightarrow \text{BWM} \rightarrow (a_t, \hat{s}_{t+1}, \mathcal{G}_{t+1})$; the first step after initialization or reset is unfiltered.

The mixture components represent distinct behavior modes. For example, from standing, walking commands at different speeds may occupy one region and jumps of different heights another. We first identify which behavior mode the user most likely intends. Specifically, we select the mode with maximum posterior responsibility under the standard GMM:

$$k^* = \arg \max_k \left[\log \pi_k - \frac{1}{2} \sum_{i=1}^{d_z} \left(\left(\frac{z_i - \mu_{k,i}}{\sigma_{k,i}} \right)^2 + 2 \log \sigma_{k,i} + \log 2\pi \right) \right]. \quad (3)$$

Let $M_k^2(z) = \sum_{i=1}^{d_z} ((z_i - \mu_{k,i})/\sigma_{k,i})^2$ denote the squared Mahalanobis distance to component k . For the selected Gaussian, $M_{k^*}^2(z) = R_{\text{safe}}^2$ is an equal-density contour; we treat its interior $M_{k^*}^2(z) \leq R_{\text{safe}}^2$, where density exceeds this threshold, as the admissible region of behaviors likely feasible in the current environment. A command inside this region passes unchanged; otherwise, we retract it toward the selected mode:

$$z_t^* = \mu_{k^*} + \sqrt{R_{\text{safe}}^2 / M_{k^*}^2(z_t^{\text{raw}})} (z_t^{\text{raw}} - \mu_{k^*}). \quad (4)$$

This radial rescaling reaches the ellipsoid boundary along the requested direction, retaining as much intent as the selected mode permits. We use the Mahalanobis radius $R_{\text{safe}} = 3$ (thus $R_{\text{safe}}^2 = 9$) and analyze its sensitivity in Figure 5.

TABLE II

GENERALIST PERFORMANCE. SIM-TO-SIM COMPARISON ACROSS STANDARD, TERRAIN, IMPLAUSIBLE-COMMAND, AND FALL BENCHMARKS.

Method	Standard			Terrain			Implausible		Fall	
	MPKPE↓	RootVel↓	SR↑	MPKPE↓	RootPos↓	SR↑	MPKPE↓	SR↑	SR [†] ↑	Jerk↓
SONIC [3]	82.3	189.6	94.1	331.2	294.7	15.3	327.6	50.0	5.9	1295.5
HoloMotion-1 [4]	109.4	121.3	89.0	330.0	280.7	18.7	248.7	67.6	0.7	2000.0
Humanoid-GPT [5]	90.9	205.8	91.9	283.3	326.7	14.0	208.0	70.6	2.9	3598.1
InterTrack (ours)	76.6	211.1	96.3	93.3	100.7	81.3	158.0	83.1	99.3	1050.6

Note: Position errors are in mm, RootVel in mm/s, Jerk in rad/s³, and SR in %. [†]Fall SR is recovery rate; all other SR columns treat a fall as failure. Each method is rolled out to the end of every reference.

We additionally train physical robustness: fallen-state initialization enables fall recovery, while persistent random external forces at the wrists and torso prepare the policy for object interaction. At deployment, the command filter neither blindly executes an implausible command nor resorts to an emergency stop, but produces a stable best-effort behavior.

IV. EXPERIMENTS

We show that InterTrack acquires interaction capabilities and robustness while also improving free-space tracking performance. We further evaluate the accuracy of the automatic terrain-generation pipeline and conduct ablations of the behavior world model and terrain-annotation modules.

A. Experimental Setup

a) *Benchmarks and metrics:* **Standard** contains 136 clips from the AMASS [38] test split. **Terrain** contains 150 held-out Bones-Seed [34] interaction clips with their annotated geometry. **Implausible** contains 136 MotionMillion [35] clips with self-penetration, missing supports, or motions incompatible with the robot morphology. **Fall** uses the Standard references but initializes the robot in fallen poses. Mean per-keypoint position error (MPKPE) measures tracked-link error; RootPos and RootVel measure root position and linear-velocity error; joint jerk measures recovery smoothness. Standard, Terrain and Implausible SR require tracking without falling. Fall SR requires the robot to stand up and resume tracking. All clips run to completion, including failed trials, avoiding survivorship bias in error metrics. Further evaluation criteria are provided in Appendix B.

Metrics are selected by the question posed by each split. Standard uses root velocity because absolute position on an unstructured floor is arbitrary, whereas Terrain uses root position because a few centimeters of drift can place a foot on the wrong side of a step. For the Implausible split, tracking-error magnitude is secondary because the references are physically implausible: we report only MPKPE to coarsely measure retained intent, while SR is primary and here means survival without dangerous collisions or falls. Fall begins in a failed state, making ordinary SR undefined, so we report successful return to tracking and jerk. The latter separates a controlled recovery from high-frequency motion that may stand in simulation but is unsuitable for hardware.

b) *Baselines and protocol:* We compare with three leading real-time general-purpose whole-body trackers: SONIC [3], HoloMotion-1 [4], and Humanoid-GPT [5],

using their official checkpoints. For fairness, we restrict baselines to general-purpose whole-body trackers supporting low-latency teleoperation, excluding methods such as PGM [17] that enable interaction via long future horizons at the expense of real-time responsiveness. Every policy is evaluated sim-to-sim in MuJoCo [31]. InterTrack uses its complete controller, including the filter with $R_{\text{safe}} = 3$. Hardware deployment runs at 50 Hz with a 0.2 s look-ahead buffer for ten future reference frames.

B. Generalist Performance Across Regimes

While supporting terrain and object interaction and robust fall recovery, InterTrack maintains strong flat-ground tracking, achieving lower Standard MPKPE (76.6 mm) and higher success rate (96.3%) than all three evaluated tracking baselines. We also evaluate real-robot performance on various demanding motions, including squat jumps, high kicks, and Tai Chi stances. Results are detailed in Appendix C.

This advantage becomes more pronounced on Terrain: these baselines reach only 14.0–18.7% SR and 283.3–331.2 mm MPKPE, whereas InterTrack reaches 81.3% SR and 93.3 mm MPKPE. Simply scaling up conventional flat-ground training with randomization therefore does not naturally generalize to terrain interaction. In contrast, InterTrack learns the distinct environment-conditioned dynamics across terrains.

Beyond terrain interaction, InterTrack is also highly robust to both physically implausible commands and falls. Under physically implausible commands, InterTrack survives 83.1% of clips and stays closest to the requested motion, producing a “best-effort” response that preserves as much of the original intent as possible without interrupting the task. The same policy thus falls back gracefully when a command exceeds its capabilities, distinguishing deliberate adaptation to capacity limits from accidental tolerance. From fallen states, it recovers in 99.3% of trials without switching controllers, compared with 0.7–5.9% for the baselines. It simultaneously achieves the lowest average joint jerk, outperforming all three evaluated baselines on both recovery metrics.

C. Environment Interaction

The matched hardware trials in Figure 4(a,b) compare terrain interaction with the same live operator command stream broadcast simultaneously to both robots. SONIC fails to exploit the terrain for meaningful interaction: it only half-squats without sitting and raises a leg without using

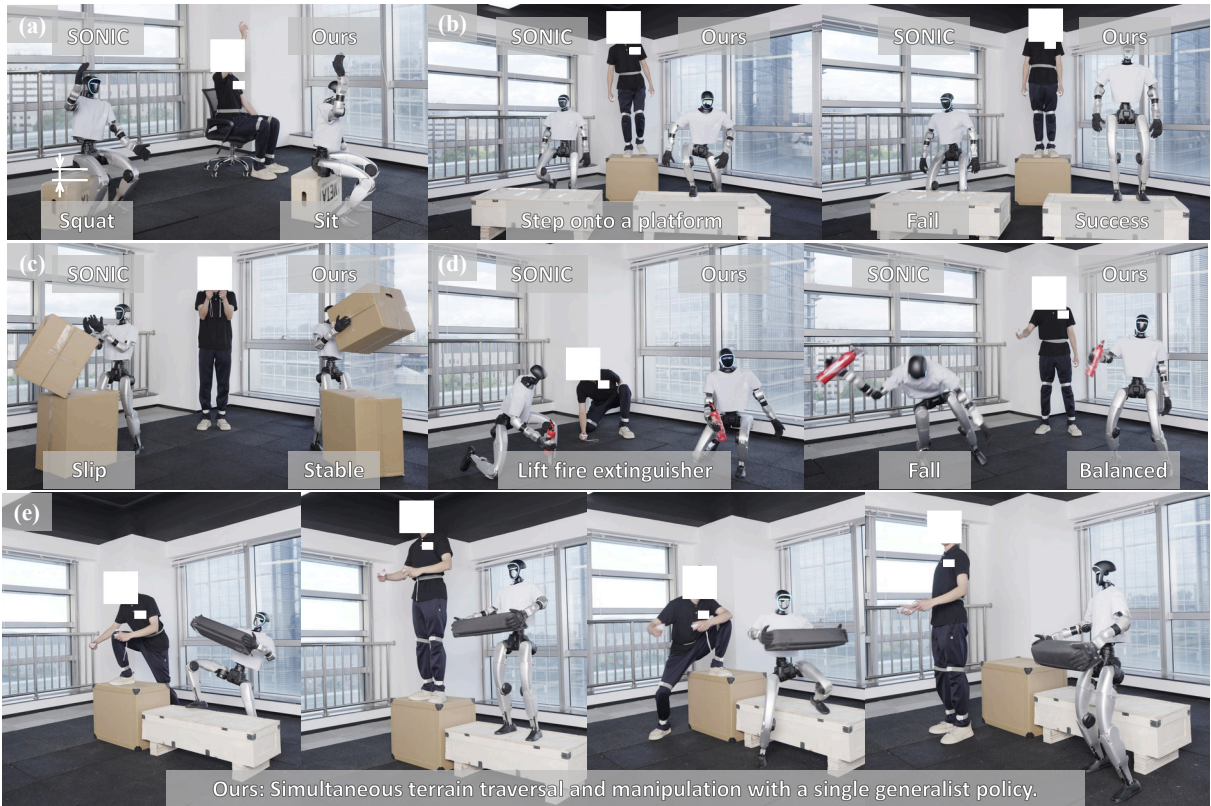


Fig. 4. **Environment interaction on hardware.** The same live operator command stream is broadcast simultaneously to the two robots running SONIC (left) and ours (right). (a,b) Terrain interaction: sitting on a box and stepping onto a platform. (c,d) Object interaction: lifting a box and rising with a fire extinguisher. (e) Unified interaction: ours carries a case while stepping onto and down from a platform.

the platform to step up, whereas InterTrack completes both motions. Panels (c,d) compare object interaction. In (c), InterTrack applies the forces required to lift and hold the box, whereas SONIC merely imitates the lifting motion and drops it. In (d), InterTrack captures and adapts to the payload-induced dynamics of the fire extinguisher, lifting it stably, whereas SONIC loses balance and falls. Panel (e) further shows that the same policy can simultaneously perform cross-terrain mobility and object manipulation: it carries a case across a platform in one continuous sequence without switching controllers.

D. Robustness

The hardware sequences in Figure 6(a,b) remove supports assumed by the command. Rather than commit to nonexistent contact or freeze, the policy preserves the achievable part of the motion. The same tracker stands up after being knocked down in panel (c). Panels (d,e) present a qualitative ablation of the command filter in simulation: without it, the controller attempts a self-interfering kick and falls (though later gets back up autonomously); with it, the controller makes a safe best-effort response—turning as commanded while suppressing the destabilizing kick.

We further sweep the safety radius to quantify the filter’s precision–robustness trade-off in Figure 5. We use $R_{\text{safe}} = 3$ throughout as a robustness-oriented trade-off. Tightening the radius to $R_{\text{safe}} = 1$ increases Standard MPKPE by 46.7 mm for just 2.2 more Implausible SR percentage points. For

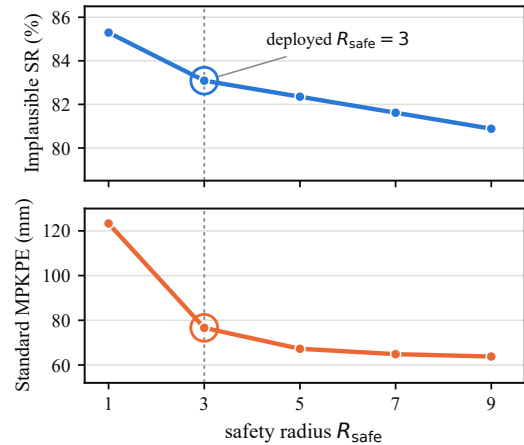


Fig. 5. Standard-split MPKPE and Implausible SR as a function of the safety radius. $R_{\text{safe}} = 3$ is used in the main experiments.

$R_{\text{safe}} > 3$, Standard MPKPE is comparatively flat, whereas Implausible SR remains visibly responsive to the radius.

E. Ablation Study

Table III treats the contents of Sections III-B to III-D as the three components. *w/o Sec. III-B BWM* removes the core behavior-world-model mechanism in Section III-B: future predictions and their associated objectives. It also removes the robustness module in Section III-D, including the prediction-dependent command filter and fallen-state initialization. *w/o Sec. III-C Terrain Data* retains the BWM and robustness mechanisms but removes the annotation module



Fig. 6. **Robustness to invalid supports, disturbances, and commands.** (a,b) With an expected chair or platform absent, the policy settles into a stable best-effort motion. (c) The tracking policy recovers after a kick-induced fall. (d,e) For the same physically implausible spinning-kick reference, filtering suppresses the destabilizing portion while retaining the commanded turn.

TABLE III
COMPONENT ABLATIONS. EFFECTS OF THE BWM, TERRAIN DATA, AND ROBUSTNESS MECHANISMS ACROSS ALL BENCHMARKS.

Variant	Components			Standard			Terrain			Implausible		Fall	
	BWM (III-B)	Terrain (III-C)	Robust. (III-D)	MPKPE↓	RootVel↓	SR↑	MPKPE↓	RootPos↓	SR↑	MPKPE↓	SR↑	SR↑	Jerk↓
w/o Sec. III-B BWM	✗	✓	–	130.3	184.3	80.1	102.9	120.4	76.7	236.3	64.7	9.6	295.3
w/o Sec. III-C Terrain Data	✓	✗	✓	75.2	210.5	97.8	220.9	311.7	12.0	156.2	83.8	100.0	1033.7
Complete InterTrack	✓	✓	✓	76.6	211.1	96.3	93.3	100.7	81.3	158.0	83.1	99.3	1050.6

Note: Each component column represents the module described in the corresponding method subsection; “–” denotes joint removal of the robustness module with the BWM, reflecting the filter’s dependence on BWM predictions. Bold marks the best and near-best values relative to the overall spread.

in Section III-C: no terrain is generated for training. All evaluation metrics follow the definitions in Table II.

Jointly ablating the BWM and robustness mechanisms (Sections III-B and III-D) reduces recovery from 99.3% to 9.6% and raises MPKPE across all three regimes, demonstrating the integrated approach’s contribution. The recovery loss is consistent with fallen-state initialization exposing the policy to recovery scenarios, while the pronounced reduction in Implausible SR is consistent with the command filter suppressing implausible commands. The increased MPKPE suggests a benefit from predictive supervision, with next-state prediction providing a plausible mechanism through learning interaction dynamics. Although this ablation does not isolate individual contributions, its results support the proposed component-level interpretations.

Removing Terrain Data substantially degrades every metric on Terrain, with SR dropping from 81.3% to 12.0% and MPKPE increasing from 93.3 mm to 220.9 mm. This result shows that incorporating our automatically annotated terrain data into training markedly improves terrain interaction performance. The safety-radius sweep in Figure 5 quantifies sensitivity to the filter threshold, while the spinning-kick comparison in Section IV-D provides a qualitative filter ablation, and, for brevity, additional relevant experiments are presented in Appendix C.

V. CONCLUSION

We introduced InterTrack, a behavior world model for highly robust whole-body tracking with environment interaction. Jointly predicting actions, future states, and future behaviors lets the policy adapt to contact-conditioned dynamics and produce stable “best-effort” behavior under implausible commands. An automatic spatial terrain annotation pipeline supplies paired motion–terrain data at scale for training. InterTrack combines terrain and object interaction and robust fall recovery with strong flat-ground tracking, achieving lower Standard MPKPE (76.6 mm) and higher success rate (96.3%) than all three evaluated tracking baselines. It reaches 81.3% success on terrain interaction (4.3× the best real-time general-purpose baseline) and 83.1% survival under physically implausible commands, while raising fall recovery rate from single-digit baseline rates to 99.3%. Hardware experiments further demonstrate agile teleoperation, terrain and object interaction, and recovery from disturbances. These capabilities share one policy and a live command interface. Two limitations remain: the command filter recognizes unfamiliar behavior rather than physical risk, and the contact-based annotation is suited to static terrain but not finite-mass movable objects and reconstructs touched supports rather than complete scenes. Extending the reconstruction to non-supporting geometry, and connecting the world model’s state

prediction to a more direct notion of physical risk, are natural next steps.

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OpenAI ChatGPT was used to assist with implementing the methods described in the Method section and monitoring experimental progress for the Experiments section. All AI-assisted outputs were reviewed and verified by the authors, who take responsibility for the final manuscript.

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APPENDIX

A. Policy Interface and Model Architecture

The controller operates the 29 actuated degrees of freedom of a Unitree G1 at 50 Hz. Its proprioceptive input $\mathbf{s}_t \in \mathbb{R}^{67}$ contains projected gravity, base angular velocity, torso linear acceleration, and joint positions and velocities. The policy also receives the previous PD target $\mathbf{a}_{t-1} \in \mathbb{R}^{29}$ and outputs the next target $\mathbf{a}_t \in \mathbb{R}^{29}$. The reference window contains ten future frames sampled at 0.02 s intervals. During live tele-operation, buffering the incoming operator stream introduces a fixed 0.2 s look-ahead delay. The policy has been verified to run at 50 Hz on hardware. At each frame, it contains 29 joint positions, the 6D reference-root rotation relative to the current robot root, frame-to-frame root translation, reference-root translation relative to the robot, and gravity expressed in the reference-root frame, giving $\mathbf{c}_t \in \mathbb{R}^{440}$.

The FSQ reconstruction branch uses 32 levels per latent dimension. The core input $\mathbf{e}_t = [\mathbf{s}_t, \mathbf{a}_{t-1}, \mathbf{z}_t] \in \mathbb{R}^{160}$ is projected to width 256, and the Transformer uses four attention heads.

B. Implementation Details

We optimize with PPO [30] in Isaac Lab [37] over the corpus mix described in Section III-C. Training samples terrain-annotated motion with probability 0.2 and flat-ground motion otherwise. In Equation (2), $\lambda_{\text{recon}} = 0.01$, $\lambda_{\text{cycle}} = 1$, $\lambda_{\text{state}} = 0.02$, and $\lambda_{\text{beh}} = 0.01$; next-step terms are masked across episode boundaries. Each sequence-level update recomputes the current rollout segment with gradients while attaching the detached per-layer KV prefix saved at rollout start, so the attention context during optimization matches the context under which actions were sampled.

Command robustness comes from the filter in Section III-D, whereas fall robustness is trained directly within the tracker by starting a fraction of flat-ground episodes from synthetic fallen poses under a curriculum over trunk inclination up to fully prone. A validated whole-body collision proxy places the robot at ground contact, and precise tracking rewards are gated by a smooth target-relative recovery gate so the policy is not penalized too hard while standing up. Separately, persistent random external forces at the wrists and torso emulate payloads and contact loads.

Each tracking term maps an error $\mathbf{e}$ to

$$r = \exp(-\|\mathbf{e}\|^2/\sigma^2). \quad (5)$$

Table IV lists the reward weights and kernel widths, while Table V reports the randomization ranges and resampling schedules. Anchor terms act on the root, relative-body terms on tracked links in the root frame, and local terms on the head proxy, wrists, and feet in the root frame, matching the points available to a VR operator. The critic is discarded at deployment and uses a separate MLP over 1756-dimensional privileged observations: multi-frame reference targets, root pose differences, body poses, and ten frames of proprioception and action history.

For Standard, Terrain, and Implausible, success rate (SR) uses a deviation-from-reference criterion: a rollout fails if

TABLE IV

REWARD TERMS AND WEIGHTS. TRACKING TERMS USE THE GAUSSIAN KERNEL IN EQUATION (5); OTHER TERMS ARE DIRECT PENALTIES.

Reward term	Weight	σ
Anchor position	1.0	0.30
Anchor orientation	0.5	0.40
Relative body position	1.0	0.30
Relative body orientation	1.0	0.40
Body linear velocity	1.0	1.00
Body angular velocity	1.0	3.14
Local head/wrist points	2.0	0.10
Local wrist orientation	2.0	0.30
Local feet position	1.0	0.15
Action rate	$-1e-1$	–
Joint limit	-10.0	–
Undesired contacts	$-1e-1$	–
Anti-shake angular vel.	$-5e-3$	–
Ankle joint acceleration	$-2.5e-6$	–

TABLE V

DOMAIN-RANDOMIZATION RANGES AND RESAMPLING SCHEDULES.

Randomized quantity	Range	Resample
Static friction	[0.3, 1.6]	startup
Dynamic friction	[0.3, 1.2]	startup
Restitution	[0.0, 0.5]	startup
Wrist/torso mass	$\times [0.8, 2.5]$	startup
Torso CoM x	± 0.025 m	startup
Torso CoM y, z	± 0.05 m	startup
Joint default offset	± 0.01 rad	startup
Push linear x, y	± 0.5 m/s	4–6 s
Push linear z	± 0.2 m/s	4–6 s
Push angular r, p	± 0.52 rad/s	4–6 s
Push angular yaw	± 0.78 rad/s	4–6 s
Wrist/torso force	± 5 N	2–5 s
Wrist/torso torque	± 0.3 N m	2–5 s
Root-translation mask	masked w.p. 2/3	2–5 s
Reference-window noise	± 0.05	each step

the translation or orientation of either the base or torso link deviates from the reference motion beyond the corresponding threshold, or if the robot falls. For Fall, a rollout succeeds once none of these failure conditions holds before the motion sequence ends.

C. Real-Robot Motion Tracking

Figure 7 collects real-robot results beyond the primary claims evaluated in the main paper. With the same live whole-body command interface, the policy tracks demanding flat-ground motions including squat jumps, high kicks, and low Tai Chi stances. These trials demonstrate that Inter-Track retains agile flat-ground tracking alongside its terrain and object interaction and robust fall-recovery capabilities, achieving lower Standard MPKPE (76.6 mm) and higher success rate (96.3%) than all three evaluated tracking baselines (Table II).

D. Human Audit of Terrain Annotations

The automatic pipeline in Section III-C generates annotations without supervision, so no ground-truth terrain exists for direct comparison. Geometric failures such as body–terrain penetration are already filtered during generation.

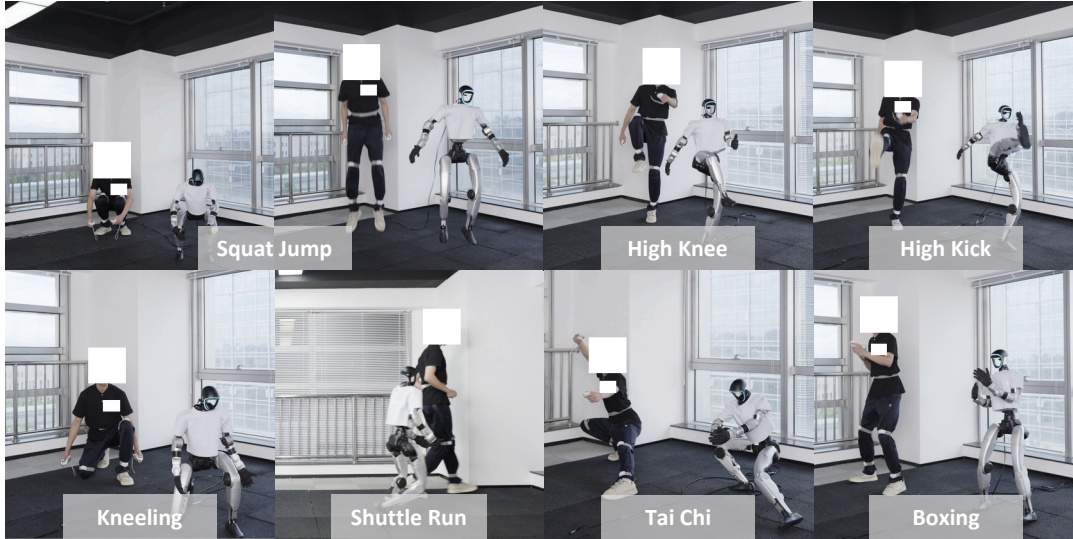


Fig. 7. **Whole-body motion capability.** One unified policy tracks live whole-body commands for squat jumps, high knees, high kicks, kneeling, shuttle runs, Tai Chi, and boxing.

TABLE VI

AUDIT OF TERRAIN ANNOTATIONS AND DOMINANT FAILURE MODES.

Terrain category	# sampled	Correct (%)	Dominant failure mode
Stairs / steps	50	92	Fast footfall obscures the deceleration signature
Chair / seat	50	94	Torso sway misplaces the backrest contact
Box / platform	50	98	Sparse contacts fail to form a primitive
Other	50	84	Hand-retargeting noise hides the velocity drop
Overall	200	92	—

What remains requires human judgment: inferring the intended interaction, assessing whether its implied contacts are physically sound, and deciding whether motion and terrain together look like something a person would actually do.

We audit 200 randomly sampled Bones-Seed annotations, 50 in each category of Table VI. An annotation is judged correct only when its reconstructed supports match the intended interaction and make the paired motion–terrain behavior plausible. Bones-Seed is used because its cleaner retargeting better isolates annotation error. Overall accuracy is 92%. Platforms are most reliable; missed short contacts account for their rare failures. Fast footfalls or torso sway can displace stair and seat supports, while wrist retargeting noise makes the hand-contact-heavy “Other” category hardest. As expected, the annotation is conservative: errors remove or misplace a support rather than introduce an intersecting obstacle.